

Lab 5

1. In the notebook, the audio waveform is transformed into time-frequency features before being passed to the keyword spotting model. What is the purpose of converting raw audio into spectrogram/MFCC-style features instead of directly using the raw time-domain waveform?

Converting audio into spectrogram/MFCC features makes speech patterns easier for the model to recognize because it highlights frequency information over time instead of raw amplitudes. This reduces noise and input size, helping small keyword spotting models train faster and perform better.

2. During full integer quantization, the notebook uses a representative dataset. What is the purpose of the representative dataset, and why is it important for creating an INT8 TensorFlow Lite model?

A representative dataset is used during quantization to show TensorFlow Lite sample inputs so it can estimate activation ranges. This helps choose accurate scaling values, reducing accuracy loss in the model.

3. What does `audio_processor.get_data()` do? Refer to `input_data.py`.

`audio_processor.get_data()` loads audio samples from the selected dataset; applies preprocessing such as time shifting, background noise mixing, spectrogram/MFCC generation; and converts them into feature tensors. It then returns the processed input features along with their corresponding label indexes for training, validation, or testing.

4. In the final model export step, the TensorFlow Lite model is converted into a C/C++ byte array. What are `g_model` and `g_model_len`, and why do they need to be copied correctly into the Arduino source file?

`g_model` is the byte array containing the TensorFlow Lite model, and `g_model_len` stores the model's size in bytes. They must be copied correctly into the Arduino source so the microcontroller can properly load and run the embedded neural network model.

5. Please include a screenshot showing the correct identification of at least one keyword, either **“YES”** or **“NO”**, as displayed in the Arduino IDE Serial Monitor after deploying your KWS model on the Arduino Nano 33 BLE.

